D. Coprime

time limit per test 3 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: brute force, greedy, number theory | https://codeforces.com/problemset/problem/1742/D

Given an array of n positive integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 1000$). Find the maximum value of $i + j$ such that a_i and a_j are coprime,[dagger] or -1 if no such i, j exist.

For example consider the array [1, 3, 5, 2, 4, 7, 7]. The maximum value of $i + j$ that can be obtained is 5 + 7, since $a_5 = 4$ and $a_7 = 7$ are coprime.

[dagger] Two integers p and q are coprime if the only positive integer that is a divisor of both of them is 1 (that is, their greatest common divisor is 1).

Input

The input consists of multiple test cases. The first line contains an integer t ($1 \le t \le 10$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains an integer n ($2 \le n \le 2 \cdot 10^5$) — the length of the array.

The following line contains n space-separated positive integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 1000$) — the elements of the array.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer — the maximum value of $i + j$ such that i and j satisfy the condition that a_i and a_j are coprime, or output -1 in case no i, j satisfy the condition.

Example

Input

```
6
3
3 2 1
7
1 3 5 2 4 7 7
5
1 2 3 4 5
3
2 2 4
6
5 4 3 15 12 16
5
1 2 2 3 6
```

Output

```
6
12
9
-1
10
7
```

Note

For the first test case, we can choose $i = j = 3$, with sum of indices equal to 6, since 1 and 1 are coprime.

For the second test case, we can choose $i = 7$ and $j = 5$, with sum of indices equal to $7 + 5 = 12$, since 7 and 4 are coprime.